Feeling the Swarm: Enhancing Drone Control with Haptics

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LAIDLAW SCHOLARS

Introduction

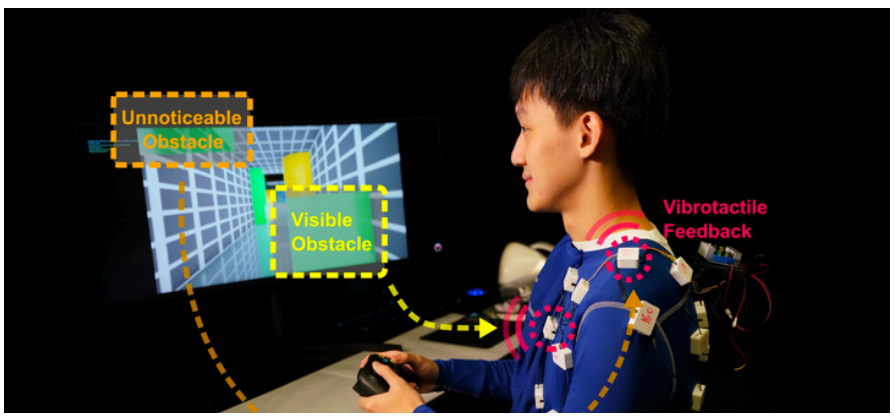
Drone Swarms: Collective Power Unleashed

Drone swarms—fleets of autonomous UAVs working in sync—are revolutionizing technology. From sweeping vast terrains in search and rescue to optimizing crops in agriculture, delivering aid in disasters, or bolstering defense surveillance, their coordinated precision and scalability redefine what's possible, pushing the boundaries of automation.



Haptics: Touching the Future

Haptics brings the power of touch to human-machine interaction, delivering tactile feedback that feels intuitive and natural. By simulating sensations like pressure or vibration, haptic systems allow users to "feel" digital environments, enhancing precision and immersion. In cutting-edge applications, haptics transforms complex control tasks into seamless, sensory-driven experiences.



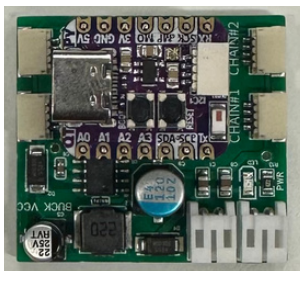
Haptics Meets Swarms: A Game-Changer

Combining haptics with drone swarms creates a groundbreaking synergy. Tactile feedback empowers operators to intuitively guide complex swarm behaviors, enhancing control in missions like search operations or tactical maneuvers. This fusion of touch and technology unlocks new levels of efficiency, precision, and impact across industries.



Devices

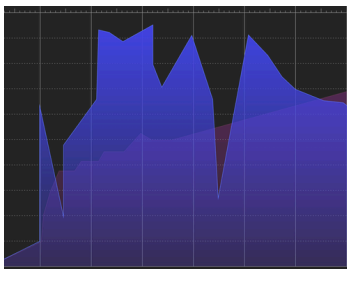
Control Unit [1]



Vibration Unit [1]



Waveform Editor



McKibben Artificial Muscle

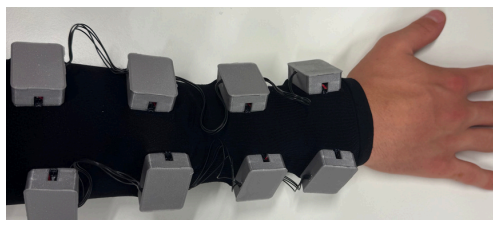


Wearble Devices

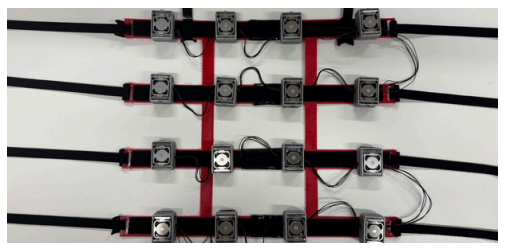
Back view



Arm Sleeve



Front View



Study: Balancing Information Clarity and Complexity

Idea: Compare & Combine three pattern types

- Buzz: Constant Vibration
- Pulse: On/Off Pulsations
- Motion: Continuous Brush Sensation

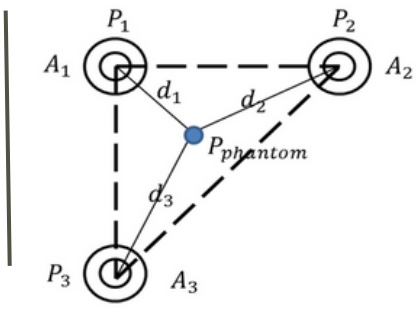
Motion Rules: [2]

Energy Summation

$$A_v^2 = \sum_{i=1}^{3} A_i^2$$

Intensity for Phantom Sensation

$$A_i = \sqrt{\frac{\frac{1}{d_i}}{\sum_{j=1}^{3} \frac{1}{d_j}}} A_v$$

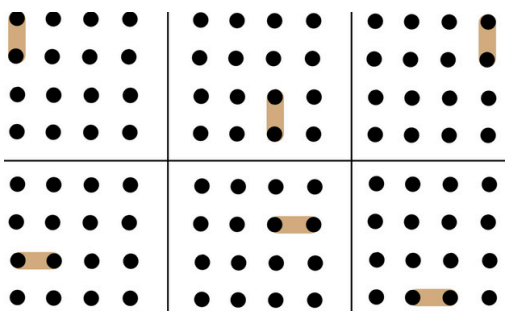


Timing for Apparent Motion

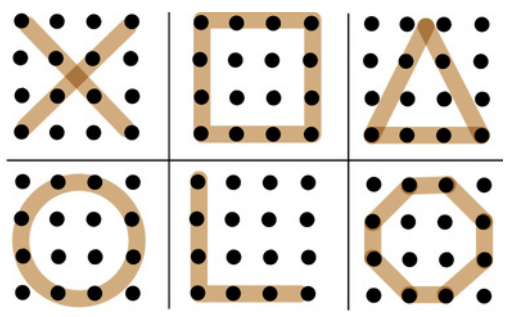
$$SOA = 0.32 \times duration + 0.0473$$

Phase A: Single pattern comparison

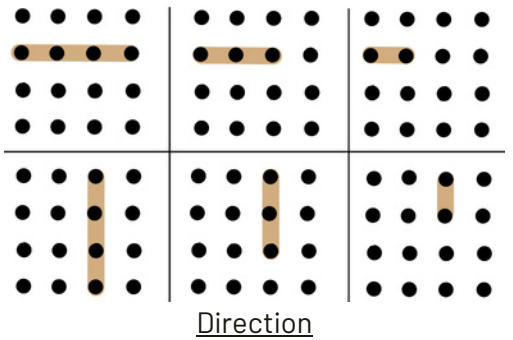
Location



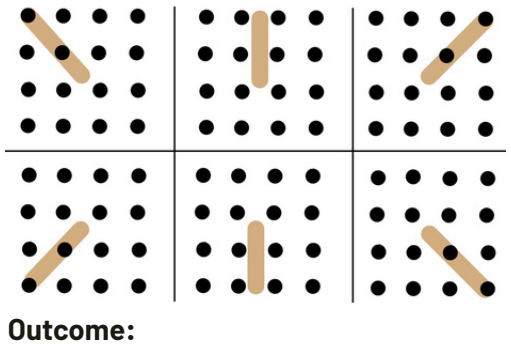
Shape



Size



Direction

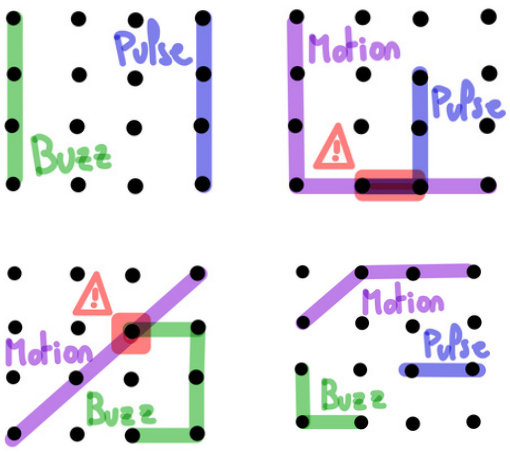


Outcome:

- Best Pattern Type for each events
- Measurements of Information Transmission per pattern

Phase B: Pattern combination

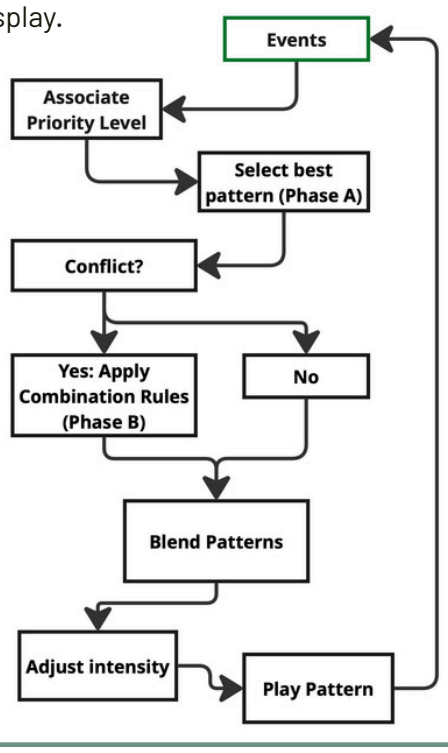
Idea: Run multiple patterns simultaneously, including conflict zones, to observe their interactions



Outcome: Patterns Combination rules

Phase C: Adaptive Haptic Orchestrator

Idea: Combine Phase A & B rules into a priority-based algorithm to optimize pattern combinations for multi-info display.



[1] Huang et al. - VibraForge: A Scalable Prototyping Toolkit For Creating Spatialized Vibrotactile Feedback Systems

[2] Park et al. - Rendering Moving Tactile Stroke on the Palm Using a Sparse 2D Array